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PII: S2095-2546(20)30037-5
DOI: https://doi.org/10.1016/j.jshs.2020.03.006
Reference: JSHS 602

To appear in: *Journal of Sport and Health Science*

Received date: 24 September 2019
Revised date: 19 November 2019
Accepted date: 14 February 2020

Please cite this article as: Angelos Vlahoyiannis , George Aphamis , Gregory C. Bogdanis , Giorgos K. Sakkas , Eleni Andreou , Christoforos D. Giannaki , Deconstructing athletes' sleep: A systematic review of the influence of age, sex, athletic expertise, sport type, and season on sleep characteristics, *Journal of Sport and Health Science* (2020), doi: https://doi.org/10.1016/j.jshs.2020.03.006

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Review

Deconstructing athletes' sleep: A systematic review of the influence of age, sex, athletic expertise, sport type, and season on sleep characteristics

Angelos Vlahoyiannis [a], George Aphamis [a], Gregory C. Bogdanis [b], Giorgos K. Sakkas [c,d], Eleni Andreou [a], Christoforos D. Giannaki [a,*]

[a] Department of Life and Health Sciences, University of Nicosia, Nicosia 1700, Cyprus

[b] School of Physical Education and Sport Science, National and Kapodistrian University of Athens, Athens 17237, Greece

[c] Department of Physical Education and Sport Science, University of Thessaly, Trikala 42100, Greece

[d] School of Sport and Health Sciences, Cardiff Metropolitan University, Cardiff CF5 2YB, UK

*Corresponding author.

Email address: giannaki.c@unic.ac.cy

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Received 24 September 2019; revised 19 November 2019; accepted 14 February 2020

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Highlights

- Several exercise- and sports-related parameters such as sex, age, athletic expertise, training season, and sports type are discussed. These factors may potentially influence sleep quantity and architecture.
- Overall, sleep quantity and quality in athletes is reduced and potentially insufficient, in comparison to the general consensus of the American Academy of Sleep Medicine for non-athlete healthy adults.
- Specifically, young athletes, mainly of Asian origin, showed poor sleep quality, with low efficiency and long wake after sleep onset. Sleep architecture data for specific athletic populations, such as female athletes, were lacking in the literature.
- Further research on athletes according to sex and age is needed in order to establish sleep recommendations and promote sleep-optimizing interventions, especially for child and adolescent athletes.

Abstract

Purpose: This systematic review aims to describe objective sleep parameters for athletes under different conditions and address potential sleep issues in this specific population.

Methods: PubMed and Scopus were searched from inception to April, 2019. Included studies measured sleep only via objective evaluation tools such as polysomnography or actigraphy. The modified version of the Newcastle-Ottawa Scale was used for the quality assessment of the studies.

Results: Eighty-one studies were included, of which 56 were classified as medium quality, 5 as low, and 20 as high quality. A total of 1830 athletes were monitored over 18,958 nights. Average values for sleep-related parameters were calculated for all athletes according to sex, age, athletic expertise level, training season, and type of sport. Athletes slept on average 7.2 ± 1.1 h/night, with 86.3% ± 6.8% sleep efficiency (SE). In all datasets, the athletes' mean total sleep time was <8 h. SE was low for young athletes (80.3% ± 8.8%). Reduced SE was attributed to high wake after sleep onset rather than sleep onset latency. During heavy training periods, sleep duration and SE were on average 36 min and 0.8% less compared to pre-season and 42 min and 3.0% less compared to in-season training periods, respectively.

Conclusion: Athletes' sleep duration was found to be short with low SE, in comparison to the general consensus for non-athlete healthy adults. Notable sleep issues were revealed in young athletes. Sleep quality and architecture tend to change across different training periods.

Keywords: Actigraphy; Polysomnography; Sleep architecture; Sleep duration; Sleep quality

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1. Introduction

Growing evidence suggests that more than half of active athletes report poor sleep during various occasions.[1] This fact bears high significance not only due to the detrimental effect of poor sleep on physical and mental performance but also because of its impact on athletes' health, as sleep is related to vital functions and may be associated with various pathological conditions and health problems.[2–4]

Sleep is a multidimensional construct that can be measured by both objective and subjective tools; thus, sleep evaluation methods may be crucial when attempting to draw safe conclusions about its impact on athletic performance.[5,6] Briefly, sleep can be subjectively assessed using specific questionnaires and sleep diaries, whilst validated objective measurements of sleep include polysomnography and actigraphic devices.[7]

An objective and detailed sleep measurement should assess the progress from wakefulness to sleep and its intermediate stages. Sleep evolves as follows: wakefulness, light sleep, deep sleep, and rapid eye movement (REM) sleep. These different states of brain activity compose the "sleep architecture". According to the American Academy of Sleep Medicine (AASM),[8] there are 4 distinct sleep stages: three are non-REM (N1, N2, and N3); and the last is the REM sleep stage. Usually, the first 2 non-REM stages (N1 and N2) are denominated as "light sleep" and the third non-REM stage as "deep sleep" or slow wave sleep.

To measure these variations in brain function, sleep electroencephalography (EEG) or polysomnography (PSG) is required. Specifically, a typical PSG study includes central, frontal, and occipital EEG, recording of eye movements, chin, and muscle activity, electromyography (EMG), electrocardiogram (ECG), pulse oximetry, respiratory effort, nasal and oral airflow, and body position sensors.[8] PSG is considered to be the gold standard method for assessing sleep and sleep disorders despite its associated practical difficulties.[7] Another critical measure for sleep is its duration, which is often referred as total sleep time (TST), while the percentage of duration of TST relative to the total time in bed is referred as sleep efficiency (SE). An additional way to estimate TST and SE is via actigraphy, which translates activity/inactivity to wake/sleep cycles,[9] and has been adequately validated against PSG in athletes.[10] Thus, actigraphy and PSG effectively measure sleep-related parameters and they are objective tools of sleep assessment. In addition, subjective methods for assessing sleep duration and quality are very popular and include various self-reported sleep logs and questionnaires. However, their validity is unclear, since it has been shown that many athletes tend to overestimate their self-reported sleep.[5,6] Subjective sleep monitoring tools may be used as initial screening tools for sleep disorders in athletes, such as obstructive sleep apnea.[11] However, PSG examination should be performed to diagnose relevant disorders,[12] and it is proposed that it could be potentially used as an indicator of poor physical recovery.[13]

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Recent reviews that have systematically examined the prevalence of sleep disorders in athletes, as well as the effect of training or competition on sleep, have included studies that both objectively and subjectively measured sleep data.[1,14] These reviews showed that athletes may experience several sleep issues[1,14] and differentiations in sleep characteristics in relation to training,[15,16] competition,[17] or other related factors, such as chronotype[16] and sports discipline.[18,19] To address these issues regarding athletes' sleep, the research community tested acute sleep hygiene strategies in order to improve athletes' sleep.[20] Nevertheless, sleep guidelines have thus far not differentiated between athletes and non-athletes,[21] making it difficult to design targeted sleep optimizing interventions. Thus, there is a need to gather more information about athletes' sleep and to develop specific guidelines for this population.[22]

Based on our literature review, quantitative information about sleep quality, sleep quantity, and architecture in athletes, using only objectively measured sleep-related data, is poorly systematized. Thus, the current review aimed to extract from the literature typical sleep characteristics of athletes according to sex, age groups, level of competence, or type of sport and summarize these data. Specifically, this review aims to:

- Provide data about objectively measured sleep quantity, quality, and architecture of athletes according to their (1) sex, (2) age, (3) athletic expertise, (4) training season, and (5) type of sport (primarily aerobic, anaerobic, or mixed sport, and individual sports *vs.* team sports).
- Address potential sleep issues and sleep fluctuations in athletes in relation to the above-mentioned categories.

2. Methodology

2.1.Information sources and search strategy

This review was performed in accordance to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.[23] Two electronic databases (PubMed and Scopus) were systematically searched by two of the authors (AV, CDG) for studies published from inception to April 2, 2019. For both databases, search terms included the following combination of keywords: ("athlete" OR "competitive" OR "sport" OR "elite" OR "player") AND ("sleep" OR "electroencephalography" OR "actigraphy" OR "polysomnography"). Retrieved records from PubMed search were limited according to article type (clinical trial, controlled trial, randomized control trial, observational study, comparative studies, and case reports) and language (English). Retrieved records from the Scopus search were limited according to source type (journal), document type (article), and language (English). A supplementary search for relevant studies was conducted from the reference list of the screened manuscripts.

2.2.Eligibility criteria

The eligibility of retrieved records for inclusion in our review was determined by two of the authors (AV, CDG) according to the following criteria: studies were written in English and measured sleep-related parameters objectively via PSG, EEG, or actigraphy, or a combination of these 3 methods. No restrictions in study design were applied. Articles were excluded if (1) participants did not take part systematically in any sport or had a history of major diseases, (2) athletic expertise or sleep-related data were inadequately described, and (3) sleep-related data were assessed after travel or any other manipulated intervention (e.g., dietary intervention, altitude, or light exposure); only the pre-intervention baseline sleep data from these studies were included, where available.

2.3.Study selection and data extraction

The process used for the literature research is depicted in Fig. 1. After removal of duplicated records, titles (and, where required, the abstracts of the remaining studies) were evaluated to select potentially relevant studies for full-text article screening. Reasons for excluding a shortlisted article during full-text screening were recorded (Fig. 1).

A standardized data extraction form was employed to obtain the following: (1) Publication details—name of first author and year of publication; (2) Participant characteristics—sample size and sex; (3) Sports-related data—type of sport and training season; (4) Sleep-related data—nights recorded, TST, SE, wake after sleep onset (WASO), Sleep Onset Latency (SOL), REM, Onset Latency (ROL), and Sleep Stages (REM, N1, N2, N3). If data were graphically presented, Plot Digitizer (Version 2.6.8; SourceForge, San Diego, CA, USA) was used to extract data of interest. Where possible, data were extracted in as many sets as possible, separating data sets for age, sex, or other sports-related parameters. The mean and SD for all data were recorded. Where required, SD was calculated from reported confidence intervals (CIs). The assumption that all groups of data were merely sub-samples of a single group was made. Thus, mean values and SDs were combined according to nights recorded.

2.4.Evidence quality appraisal

All studies were assessed for their methodological quality using a version of the Newcastle-Ottawa Scale (NOS) for cross-sectional studies (Supplementary table 1). This is an established tool, adapted by Gupta and colleagues,[1] and used in other relevant studies.[14] The NOS evaluates evidence through 8 items measuring selection,

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comparability and outcome criteria. These criteria were modified by 2 reviewers (AV, CDG) to represent adequately the aims of this review. When a criterion was satisfied, a score of 1 (or in case of Criterion 7, a score of 2) was awarded to the corresponding item. Otherwise, a score of 0 was awarded the corresponding item. The sum of the scores was used to categorize overall study evidence quality as low (<5), moderate (5–7), or high (>7).

2.5.*Definitions of sleep-related parameters and terminology*

The extracted sleep parameters are defined in Table 1. These parameters were selected for the integrative view of sleep quantity (TST), quality (WASO, SE, SOL), and architecture (ROL, REM, N1, N2, and N3).

All TST values were transformed into h for purposes of uniformity. Data in regards to sleep stages were presented in min, as measured in the majority of the studies in literature. In case of merged N1 and N2 stages of sleep (as light sleep), the results are presented but not included in the calculation of the general or subgroup mean. In the case of sleep scoring with 5 stages (REM, N1, N2, N3, and N4), the N3 and N4 stages were combined and noted accordingly.[8]

2.6.*Sample characteristics assessment and organization*

Extracted data sets were categorized according to (1) sex—males and females; (2) age—available data were categorized into 3 groups according to the participants' age: i) childhood and adolescence, 6–17 years, ii) early adulthood, 18–24 years, and iii) middle adulthood, 25–40 years; (3) sports parameters—data sets were classified according to i) sports reliance on different types of energy production pathway (primary aerobic, anaerobic, or mixed), ii) individual or team sports, and iii) the season of data collection (pre-season/tapering, in-season/competition phase, heavy training phase); and (4) athletes expertise—all participants were evaluated according to the model for classifying the validity of expert samples in sport psychology[24] to identify their level of competence (Supplemental Table 2). This tool evaluates athletic expertise by within-sports (level of performance, success, and experience) and between-sports (competiveness of the sport in the country and in the world) criteria. Due to limited data regarding athletes' expertise and success, a modified version was used that had been previously employed in other studies.[1,14] The calculated score from the modified equation (Eq. 1) was used to categorize athletes as elite (score ≥ 8) or semi-elite (score <8):[1,14]

$$A \times [(D + E)/2], \text{ (Eq. 1)}$$

where A = standard of performance, D = competiveness in country, and E = global competiveness of sport.

3. Results

3.1. Studies selection

During the literature research, 3992 studies were identified (Fig.1). After duplicate removal, 3683 records were screened by title and, where required, by abstract. A total of 159 full-text articles were assessed for eligibility. Through full-text screening, 96 articles were excluded and 18 additional records were identified, resulting in a total of 81 studies that were included in the qualitative synthesis.

3.2. Studies' characteristics

Included studies were published from 1978 to 2019, with 55.6% of the studies being published between 2016 and 2019. The sample included 1830 participants, with 230 females, 1040 males, and 560 combined (without clear sex segregation), of which 177 were children and adolescents, 1080 were young adults, 410 were middle-aged adults, while the age of 163 participants was not reported. The mean age of the participants was 22.2 ± 5.5 years. Overall, 12 of the studies evaluated sleep-related parameters using PSG, 4 studies used EEG, 61 studies used actigraphy, and 4 studies performed PSG or EEG in conjunction with actigraphy, resulting in 206 data sets (PSG/EEG n = 37, Actigraphy n = 169). A total of 18,958 recorded nights were extracted, of which 4.2% were obtained from PSG/EEG measurements and 95.8% from actigraphy. A total of 555 nights were recorded in children and adolescents, 6298 in young adults, and 2330 in middle-aged adults, while 9775 nights of the 163 participants of undefined age were excluded in this specific analysis.

3.3. Evidence quality appraisal

According to the NOS, evidence quality of 5 studies was classified as low, 56 as medium, and 20 as high (Supplemental Table 1). The first NOS evaluation component, sample selection, had a median score of 2 (0–4). The second component, comparability of studies, received a median score of 2 (0–2); and the third component, outcome measurement, had a median score of 3 (2–3).

3.4. Athletes expertise appraisal

Using the modified version of the model for classifying the validity of expert samples in sport psychology, the participants were categorized as elite (44 studies, females n = 117, males n = 716, combined males and females n = 470) or as semi-elite (37 studies, females n = 113, males n = 324, combined males and females n = 90)

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(Supplemental Table 2). Elite athletes were represented in 113 data sets of 13,262 nights recorded and semi-elite athletes in 93 data sets of 5696 nights recorded.

3.5. Overall sleep data (quality, quantity, and architecture)

Analytical sleep-related data obtained from PSG/EEG and actigraphy measures are presented in Tables 2 and 3, respectively. Data sets that included daytime naps,[25] or assessed sleep during ultra-endurance competition (duration of at least 6 h), in single[26] or multiday races,[27–29] and individual[26,28] or team ultra-endurance events,[27,29] are presented separately and only their baseline data were used for the analysis. Overall results are showed in Table 3. Athletes slept approximately 7.2 ± 1.1 h per night. SE was 86.3% ± 6.8%, with mean SOL time 14.8 ± 17.0 min and WASO 52.7 ± 32.0 min. According to PSG/EEG measures, REM stage occupied 114.1 ± 35.4 min of sleep, N1 stage occupied 47.9 ± 31.0 min, N2 occupied 240.5 ± 53.6 min and N3 occupied 91.5 ± 35.8 min.

3.6. Sleep quantity and quality according to sex

Discriminating data sets for sex showed that male athletes slept a total of 7.2 ± 1.1 h with 15.7 ± 17 min SOL, and female athletes slept 7.5 ± 1.2 h, with 9.7 ± 11.2 min SOL. SE was 85.2% ± 7.2% and 89.1% ± 4.8% for males and females, respectively. Sleep architecture data were available for male athletes only, as none of the included studies reported sleep architecture specific to female participants. Therefore, available data show that male athletes' sleep stages comprised 105.0 ± 33.8 min REM, 55.8 ± 29.0 min N1, 246.7 ± 53.3 min N2, and 82.9 ± 42.0 min N3.

3.7. Sleep quantity and quality according to athletic expertise

Classification based on athletes' level showed no differences in TST between elite and semi-elite participants. SOL was lower in semi-elite athletes than in elite athletes (9.5 ± 10.5 min *vs.* 18.4 ± 19.4 min). SE was 1.5% less in elite athletes compared with semi-elite athletes. It was found that elite athletes' sleep was composed of less N1 and N2 sleep stages and more of N3 and REM sleep stages compared with semi-elite athletes' sleep. In comparing SE between male and female elite athletes, it was found that SE was 4.6% less in males than in females. Sleep architecture data for elite athletes were very limited in the literature (N1 and N2 are presented as data from one study).[30] To the best of our knowledge, there were no discrete data according to sleep stages for male or female elite athletes and for semi-elite female athletes.

3.8. Sleep quantity and quality according to age

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As shown in Table 4, TST varied significantly throughout age groups. Children and adolescent athletes slept on average 60 min less than the young adults group, and 36 min less than middle-aged adults. No significant difference in SOL was found among age groups. SE was 5.6% and 6.6% less in children and adolescents, compared to young and middle-aged adults, respectively. The same trend was present for WASO. According to sleep architecture, data were lacking for middle-aged adults. REM sleep stage was 15% less in children/adolescents than in young adults, whilst there were no available data for middle-aged adults. Athletic expertise classification between age groups showed that SOL was longer in each elite age group, as compared to the semi-elite corresponding group.

3.9. Sleep quantity and quality in relation to training season

For both sleep quality and sleep quantity, moderate fluctuations were present among training seasons. TST was less during heavy training phases than in pre-season/tapering phases or competition phases (6.7 h *vs.* 7.3 h *vs.* 7.4 h, respectively). SE was found to be less during heavy training phases and greater during in-season/ competition periods (84.5% *vs.* 87.5%). According to sleep architecture, even if data are elusive, N3 sleep during the pre-season/tapering phase was reported to be longer than during the in-season/competition and heavy training phases (130 min *vs.* 97 min *vs.* 74.6 min, respectively).

3.10. Sleep quantity and quality in relation to sports type

The majority of included articles studied participants in purely aerobic or mixed sports, whilst only 2 studies[9,31] included participants in sports that require mostly power/explosiveness. According to this categorization, SOL was longer in power/explosiveness sports compare to other sports, with a large SD (24.2 ± 31.3 min). According to sleep architecture, athletes in anaerobic sports had slightly less REM sleep compared with athletes in aerobic or mixed sports (103.0 min *vs.* 111.8 min *vs.* 107.9 min, respectively). N3 “deep sleep” was longer in athletes in mixed sports than in athletes in aerobic sports, and over two-fold longer than athletes in power/explosiveness sports (139.5 min *vs.* 78.2 min *vs.* 61.0 min, respectively). Comparing individual and team sports, no great differences were present, except for N3 sleep, which was longer in athletes participating in team sports than for athletes participating in individual sports.

3.11.Sleep during ultra-endurance competitions

The majority of the studies used actigraphic devices to evaluate sleep after ultra-endurance competitions. During multi-day ultra-endurance competitions, mean TST

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ranged from 7.0 ± 0.9 h to 7.7 ± 2.0 h, with SE ranging from 83.0 ± 6.0 h to 85.1 ± 5.9 %. In case of extreme ultra-endurance team-sport races such as roparun, athletes obtained a mean of 2.2 ± 0.3 h of TST with 83 ± 6% SE. According to sleep architecture, no longitudinal EEG data were available for multi-day ultra-endurance events. Nevertheless, it was shown that immediately after an ultra-triathlon race, REM sleep decreased, wakefulness increased, and slow wave sleep, corresponding to N3 sleep stage, was no different compared to a no-exercise scenario.

4. Discussion

The current review aimed to summarize objectively measured sleep data from a wide range of athletic populations in order to (1) develop an integrative perspective of athletes' sleep, (2) identify potential sleep issues or related risk factors, and (3) reveal possible gaps in current knowledge. Since norms and recommendations for athletes' sleep have not been established yet, this review points towards a general depiction of athletes' sleep. The current review reveals that sleep quantity of athletes is reduced and potentially insufficient according to the general consensus of the AASM for non-athlete healthy adults.[32] Thus, overall sleep quality appears inadequate, which highlights the need for sleep-optimizing interventions, especially for children and adolescent athletes, for which specific guidelines are not available. Furthermore, periods of heavy training load were identified as being more sensitive to potential alterations in sleep quality, while sleep architecture may be modified during pre-season.

4.1.Sleep quantity

Evidence suggests that sleep duration in athletes is limited to 7.2 h/night, across all investigated categories. All studies reported TST mean values shorter than 8 h per night, which is considered to be at the lowest values of the cut-off point of 7 h according to the guidelines of AASM for healthy adults.[32] Notably, the shortest mean TST was 6.3 h, which was reported in children and adolescents,[33] in contrast with the current AASM guidelines for the paediatric population, which suggest 8–12 h/night.[21] In early adulthood, average sleep duration was marginally within the suggested >7 h of sleep per night.[32] In contrast, in middle-aged athletes, TST was below this recommended cut-off. However, these suggested limits for a healthy population may underestimate sleep needs in athletes, since the recovery demand in sports is increased.[34] This fact is supported by studies that found improvements in sports performance when sleep was extended up to approximately 2 h.[3,4]

The prevalence of insufficient sleep duration in elite athletes is well documented in the literature.[35] In most cases athletes appear to obtain less than 8 h of sleep per night.[36] This issue is also highlighted in young athletes.[33,37] Student-athletes were found to sleep less that their non-athletic counterpart population.[38] Similarly,

adolescent athletes had less sleep than senior athletes.[39] However, these studies included mainly athletes of Asian origin; hence, these results could have been influenced by the habits of this specific ethnic group.

Several factors, such as training or competition, have been shown to consistently correlate with athletes' sleep quantity.[14] An athlete's training schedule impairs sleep duration because it reduces the time available for sleep.[40,41] For young athletes, their daily schedule, including school attendance, may be a key factor in reducing sleep quantity. Furthermore, reduced sleep duration is often observed during training camps,[42,43] where not only training schedules but training volume and intensity are increased. It has been previously shown that sleep duration is impaired in endurance athletes who experienced overreaching due to high training volume [28,44] and that high-intensity training requires sufficient sleep time since recovery demands are elevated.[45] For example, during multiple-day ultra-endurance competitions, such as roparun (a non-stop relay run between Rotterdam and Paris), athletes sleep is considerably reduced to 2.2 h.[29]

Because sleep is a key restorative factor for daytime-induced fatigue, insufficient sleep duration may be a risk factor for several hormonal imbalances that suppress muscle growth[46] and may result in unfavourable body composition.[47] Inadequate sleep quantity in athletes extends its consequences to sports performance. In contrast, benefits derived from adequate sleep duration translated into higher team rankings in a netball tournament.[48] Overall, variation in sleep quantity affects mostly psychomotor vigilance.[49] Because adolescent athletes tend to sleep more over the weekends to counterbalance their weekday sleep debt, their psychomotor vigilance time was found to be shorter on Monday than on Tuesday or Friday.[50] Additionally, in a study by Choi and colleagues,[33] longer sleep duration correlated with improved shuttle bouncing performance in junior badminton players.

4.2.Sleep quality

A key finding of the present review is that reduced sleep quality is reported among athletes. In the reviewed athletic sample, children and adolescents, independently of athletic expertise, exhibited a notably low SE (80.3%). However, SE alone cannot recognize short episodes of wakefulness; therefore, examination of SOL and WASO is important. In the present review, no significant differences were reported for SOL, but WASO was longer among children and adolescents (74 min). This fact may be indicative of long or frequent wake bouts during nocturnal sleep. Furthermore, low SE and elevated WASO was more noticeable in pre-season training compared with in-season and off-season (Table 4).

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Overall, there seems to be an interrelation among several sleep quality parameters such as SOL, WASO, and SE.[9] As shown in elite young athletes/rowers, falling asleep quickly (reduced SOL) was related to more restful sleep and fewer awakenings.[37] However, overall SE was shown to be lower in elite athletes compared to the age- and sex-matched non-athletic population.[9]

As with sleep quantity, sleep quality is affected by training and competition.[51] A short period of intensified training results in progressive decrements in sleep quality among cyclists and triathletes.[44,52] Extending this period—for example, during training camps[42] or during a cycling grand tour[28]—may also impair SE. Overall, Wall and colleagues[53] suggested that SE may be an objective predictor of overreaching in swimmers.

In regards to other health issues, low SE values are indicative of the development or establishment of health problems such as increases in systolic blood pressure[2] or lower cognitive function.[54] Also, Choi and colleagues[2] showed that higher SE was related with better shuttle bouncing performance in young badminton players.

4.3.Sleep architecture

The data we reviewed on sleep architecture showed that athletes of both sexes spend approximately 114.1 min of their sleep time in the REM stage, 47.9 min in N1 stage, 240.5 min in N2 stage, and 91.5 min in N3 stage. Converting these sleep stages from min to percentages of TST, the following distribution is obtained: 23% REM, 9.7% N1, 48.7% N2, and 18.5% N3 sleep stage. In comparison, a typical non-athletic adult spends approximately 25% of sleep time in REM stage, 5% in N1 stage, 50% in N2 stage, and 20% in N3 stage.[8] Early studies showed that athletes tended to spend more time in non-REM sleep and less time in REM sleep compared to non-athletic controls.[55] However, ultra-endurance competition changes athletes' sleep architecture by increasing REM sleep stage, while N3 sleep stage is not affected.[26]

A factor that was identified as being associated with alterations in sleep architecture in athletes was pre-season training period. It was observed that during the pre-season period the amount of sleep in the N3 sleep stage is extended. However, these data were calculated from a limited number of studies, and generalized assumptions should be made with caution.

Findings based on training volume and sleep architecture show that after endurance races, REM proportion is reduced and N3 stage sleep may increase.[26,56] Driver and colleagues[26] found that the REM stage was a more sensitive indicator of exercise-induced stress than N3, since REM changed only after 15 km and 42.2 km runs. Sapiro and colleagues[56] showed that after a 92 km road race, REM proportion was decreased and N3 stage sleep was increased. During pre-season and peak training period, N3 sleep has also been found to increase.[57] Recent data show that N3 stage

sleep increases in response to higher physiological restorative demand in athletes (as indicated by higher resting heart rate and lower heart rate variability).[58] Trinder et al.[31] found that, according to the type of sport, N3 was higher in athletes practicing an aerobic sport compared to an anaerobic sport.

4.4. Sleep optimization interventions

Several interventions have been implemented in order to improve athletes' sleep, such as bright-light exposure,[59] cold-water immersion,[60] napping,[61] nutritional interventions,[3] and sleep hygiene education.[62] Some interventions may improve sleep in the short term, but these improvements are not sustained in the long term.[62] Some factors that were thought to negatively affect sleep, such as the use of electronic devices in the evening, should perhaps be reconsidered before relevant interventions are established.[63] Thus, awareness should be raised when investigating easily applicable and targeted sleep optimization interventions.

Supplementing nocturnal sleep with daily naps has shown mixed results. In adolescent athletes, daily naps showed varying effects on sports-specific performance. For example, 20-m run performance improved but no effect was observed in shooting performance.[64] Furthermore, after travel, napping showed no significant benefit in Wingate performance.[61] On the other hand, Blanchfield and colleagues,[65] reported benefits in endurance performance after nap supplementation of nocturnal sleep among athletes with TST <7 h. Similarly, Waterhouse and colleagues[66] showed that post-lunch napping improved alertness and sprint performance in sleep-deprived subjects.

More promising results were found in studies that tried to prolong nocturnal sleep duration. In basketball players, extension of TST correlated with improvements in athletic performance.[4] In a recent study, extension of sleep duration induced by a meal containing high glycemic index carbohydrates was related to faster reaction time the following morning.[3] Notably, sleep quantity and quality were modified by admininstering a post-exercise meal containing high glycemic index carbohydrates, thus underlining the need for easily applicable sleep-optimizing interventions. For example, tart cherry juice was found to be an effective nutrition intervention for improving sleep duration and quality in healthy individuals.[67] Nutritional interventions should be used with caution, however, as they may result in improved athletic performance but still have a negative effect on sleep. For example, a study by Miller and colleagues[68] showed that caffeine supplementation lead to improvements in endurance sports performance but impaired sleep duration in athletes.

4.5. Other factors that may affect sleep in athletes

Sleep is a multidimensional construct that alters throughout the human life span according to an individual's habits, such as electronic device use.[69] Due to the nature of both their training and competition, athletes are exposed to several additional risk factors that may alter sleep patterns and induce sleep deprivation. For instance, environmental factors that influence sleep include traveling across different time-zones,[70] training in heat,[60] conditions that lead to hypoxia,[71,72] and psychological issues derived from competition stress.[73]

Practicing a sport since adolescence seems to alter sleep quantity. It has been shown that sleep onset and wake time differ between athletes and non-athletes, with young athletes going to sleep earlier and waking earlier compared to controls who are training recreationally or are sedentary.[74] Because training schedules demand that student athletes wake up early in the morning, electronic device usage may contribute considerably to the reduced sleep of adolescent athletes. For example, the use of smartphones after lights out may impair athletes' sleep,[69] and unrestricted Internet access appears to be related to fewer hours of sleep among young athletes.[75] On the other hand, some recent studies have shown that electronic device use does not necessarily affect sleep quantity and quality in young adults.[6,63]

Recently, Fowler and colleagues[70] showed that professional rugby players who travelled from Australia to the UK needed on average of 5 days to restore their sleep-wake patterns. Waterhouse and colleagues[76] showed that appropriate itinerary arrangements may reduce some of the negative effects of jet lag. In contrast, one study showed that interstate travel did not affect sleep patterns in athletes on the night before a game.[77]

Pedlar and colleagues[71] found that sleep duration was not impaired when training was performed at high altitude and that sleeping in a normobaric hypoxic tent led to impaired duration of stage N1 and N2 sleep but led to no significant changes in deep sleep or REM sleep. Sleeping at high altitude showed that REM sleep initially falls acutely but afterwards rises gradually from day 1 to day 15.[78,79] In contrast, deep sleep was shown to decline gradually during exposure to altitude.[79] Similarly, sleep disturbances as a response to high altitude have also been observed in female middle distance runners and young soccer players of various levels.[78,80,81] Ambient temperature is an additional environmental parameter that may impair sleep. As shown by Skein and colleagues,[60] 5 consecutive days of training in the heat can impair sleep quality.

Another factor that has been investigated for its effect on athletes' sleep is chronotype. It is proposed that chronotype not only affects ratings of perceived exertion and fatigue scores and performance,[16] but it also affects sleep quality following exercise training.[82] In regard to chronobiology and circadian rhythm, it was recently shown that rest–activity circadian rhythms vary among different sport disciplines,[18] and this has an impact on sleep onset and wake times.[18]

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4.6.Methodological issues and future research directions

The current study aimed to analyze athletes' sleep intra-variability using only objectively measured sleep-related data while excluding extreme environmental factors that could affect sleep. Objectively measured data was used because it is well documented that subjective measures, such as sleep logs, often overestimate athlete sleep duration.[5,6] Likewise, commercially available wearable devices also show poor accuracy and reliability for sleep tracking.[83,84] In athletic populations, wearable devices may overestimate TST, providing questionable results.[10] Nevertheless, wearables are practical, user-friendly, non-invasive devices that, after adequate validation, could be useful for coaches and sports scientists to use in longitudinally monitoring athletes' sleep.

Analyzing only objectively measured data revealed gaps in current methodologies and existing literature. When comparing data across studies, differences in the way each sleep variable was calculated may significantly affect the outcomes. For example, in some studies,[27,81] SE was artificially higher because its calculation did not include the SOL measure. Regarding sleep-related measures, data were scarce for certain categories of participants, and especially for sleep architecture measures. Sleep stages of female athletes, at both the elite and semi-elite levels, were not presented separately. Sleep architecture data were also understudied for elite male athletes and middle-aged adult athletes. No data were retrieved in the literature search for masters athletes or athletes in late adulthood.

Thus, it is difficult to measure sleep architecture in elite athletes across all ranges of the human lifespan. Therefore, additional studies are needed to investigate sleep stages and their relations to various physiological measures in athletes. Future studies should also focus on sleep evaluation of female athletes. An important outcome of this review is that athletes, especially children and adolescents, present overall poor sleep quality and quantity. Thus, easily applicable interventions should be investigated to optimize sleep and sleep-related parameters in junior athletes.

5. Conclusion

The current systematic review showed that athletes slept on average 7.2 h per night with 86.3% SE. Notably, athletes in junior categories demonstrated low SE (80.3%), with no significant increases in SOL, but with long WASO. Several exercise- and sports-related parameters that can potentially influence sleep quantity and architecture were analyzed, such as athletic expertise, training season, and sports type. Data were lacking for certain age- and sex-specific categories, such as females and masters athletes. Therefore, more studies should be conducted to evaluate sleep architecture among these categories of athletes. Further research is needed to establish sleep

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recommendations for athletes and promote sleep-optimizing interventions, especially for children and adolescent athletes.

Authors' contributions

AV performed the literature review and data extraction and wrote the first draft of the manuscript; CDG performed the literature review and data extraction, and wrote and revised the first draft of the manuscript; GA, GCB, GKS, and EA revised the manuscript. All authors have read and approved the final version of the manuscript, and agree with the order of presentation of the authors.

Competing interests

The authors declare that they have no competing interests.

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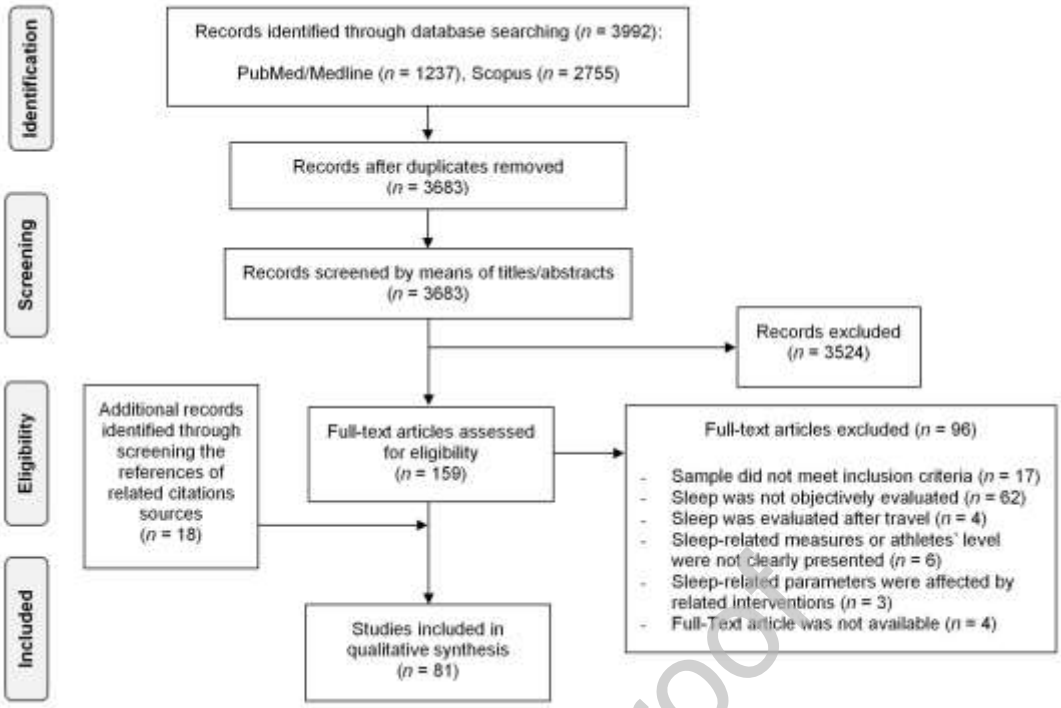

Fig. 1 The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow chart for study selection.

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Table 1

Definition of sleep-related terms used in data extraction.

Term	Definition	Estimated by polysomnography/ electroencephalography	Estimated by Actigraphy
Nights recorded	The number of the nights that sleep-related parameters were evaluated, multiplied by each participant.	+	+
TST	The total amount of time whilst asleep in bed, presented in h.	+	+
Sleep efficiency	TST expressed as percentage of total time in bed. Sleep efficiency is used as an indicator of sleep quality, and a value <85% is indicative of sleep disorders. (TST / Time spent in bed) × 100	+	+
Sleep onset latency	The time elapsed from laying down on the bed until N1 stage sleep onset, presented in min.	+	+
REM onset latency	The time elapsed from laying down on the bed until REM stage sleep onset, presented in min.	+	–
Wake after sleep onset	The total amount of wakefulness from N1 stage sleep onset until the final awakening, presented in min.	+	+
REM stage sleep	The stage of sleep during which REM and dreaming occurs, presented in min.	+	–
N1 stage sleep	The first sleep stage, represented in min.	+	–
N2 stage sleep	The second stage of sleep, represented in min.	+	–
N3 stage sleep	The third stage of sleep, often referred as "deep sleep", presented in min.	+	–

The combination of N1 and N2 stages of sleep, often referred as "light sleep".

Notes: The symbol "+" denotes that the sleep-related parameter can be measured by the specific sleep measure. The symbol "-" denotes that the sleep-related parameter can not be measured by the specific sleep measure.

Abbreviations: REM = rapid eye movement; TST = total sleep time.

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Table 2

Sleep-related data measured with either polysomnography or electroencephalography (mean ± SD).

First author (publication year)	*n*	Sex	Age (year)	Type of sport	Training season	Nights recorded	TST (h)	SOL (min)	ROL (min)	SE (%)	WASO (min)	REM (min)	N1 (min)	N2 (min)	N3 (min)
Walker (1978)[55]	10	M	19.3 ± NR	Distance running	Competition phase	10	7.6 ± 0.2	12.1 ± 6.2	78.4 ± 21.9	–	6.6 ± 11.3	117.0 ± 15.2	–	–	–
						20	7.7 ± 0.1	15.2 ± 6.3	90.8 ± 26.2	–	4.0 ± 2.9	113.7 ± 17.8	–	–	–
Shapiro (1981)[56]	6	M	21.7 ± NR	Running	Competition phase	6	7.1 ± NR	–	–	–	–	116.3 ± NR	18.3 ± NR	198.9 ± NR	90.5 ± NR[a]
						6	8.2 ± NR	–	–	–	–	65.5 ± NR	38.3 ± NR	175.7 ± NR	214.1 ± NR[a]
						6	8.9 ± NR	–	–	–	–	93.0 ± NR	15.7 ± NR	212.4 ± NR	212.1 ± NR[a]
						6	7.8 ± NR	–	–	–	–	141.5 ± NR	23.2 ± NR	180.8 ± NR	119.8 ± NR[a]
						6	7.5 ± NR	–	–	–	–	126.1 ± NR	20.1 ± NR	213.5 ± NR	89.0 ± NR[a]
Trinder (1982)[85]	10		22.3 ± 2.3	Distance running	NR	20	9.2 ± 1.1	17.0 ± 8.1	88.0 ± 46.2	–	–	121.0 ± 27.1	64.0 ± 30.5	278 ± 49.8	87.0 ± 26.0
	10	M	22.5 ± 2.6	Weightlifting and bodybuilding		20	8.4 ± 1.8	33.0 ± 52.0	91.0 ± 44.7	–	–	103.0 ± 34.1	71 ± 40.5	271 ± 55.1	61.0 ± 33.0
	10		21.1 ± 2.2	Multi-sports		20	8.9 ± 0.8	26.0 ± 25.5	73.0 ± 19.7	–	–	125.0 ± 28.0	61 ± 15.4	280 ± 43.8	69.0 ± 21.7
Driver (1994) [26]	8	M	30 ± NR	Ultra-triathlon	Competition phase	8	7.3 ± 0.8	9.0 ± 9.0	90.0 ± 31	–	–	–	–	–	–
						8	6.9 ± 1.0	10.0 ± 6.0	77.0 ± 26	–	–	–	–	–	–
					Ultra-endurance competition phase	8	7.6 ± 0.5	8.0 ± 7.0	81.0 ± 24	–	–	–	–	–	–
						8	10.4 ± 1.6	8.0 ± 4.0	186.0 ± 83	–	–	–	–	–	–
Taylor (1997) [57]	7	F	19.0 ± 2.0	Swimming	Competition phase	7	7.5 ± 0.5	19.3 ± 3.5	73.3 ± 37.6	–	4.4 ± 6.4	–	–	–	–
					Competition phase	7	7.9 ± 1.0	16.6 ± 5.6	64.3 ± 16.0	–	9.0 ± 13.8	–	–	–	–
					Tapering	7	7.6 ± 0.2	16.9 ± 8.8	60.7 ± 8.9	–	9.1 ± 5.3	–	–	–	–
George (2003) [86]	52	M	25.5 ± 2.4	Football	Pre-season	52	6.0 ± 0.5	21.7 ± 20.3	–	90.3 ± 6.8	–	–	–	–	–
Hoshikawa (2007)[87]	8	M	21.9 ± 1.6	Distance Running	Heavy training phase	8	7.6 ± 0.3	9.7 ± 3.4	86.6 ± 26.3	94.6 ± 3.7	22.6 ± 15.4	97.8 ± 22.3	61.3 ± 14	209.3 ± 34.1	86.1 ± 22.3
Gosselin (2009)[88]	7/4	M/F	22.6 ± 2.4	Multi-Sports	NR	11	7.6 ± 0.3	8.6 ± 7.3	65.6 ± 13.8	94.8 ± 4.3	17.0 ± 7.6	–	–	–	–
Hoshikawa (2010)[89]	7	M	21.7 ± 1.6	Distance Running	Heavy training phase	7	7.6 ± 0.3	9.3 ± 3.4	91.2 ± 24.6	94.4 ± 3.9	23.5 ± 16.4	97.6 ± 24.1	60.9 ± 15.1	208.9 ± 36.8	86.0 ± 24.1
Hoshikawa (2013)[80]	7	M	-	NR	Heavy training phase	7	7.6 ± 0.1	8.6 ± 3.3	–	94.5 ± 1.7	22.2 ± 8.5	103.1 ± 13	52.2 ± 25.0	236.9 ± 27.7	58.3 ± 22.7
						7	7.5 ± 0.2	14.6 ± 4.1	–	92.7 ± 2.9	24.7 ± 14.4	128.2 ± 28.3	56.0 ± 27.3	210.6 ± 30.4	50.0 ± 12.6
Sargent (2013)[78]	10	M	15.6 ± 0.5	Soccer	Pre-season	10	8.0 ± 0.7	18.0 ± 8.0	–	86.0 ± 5.0	64.0 ± 39.0	107.0 ± 25.0	26.0 ± 13.0	216 ± 28	130.0 ± 19
Sargent (2016)[10]	16	M	19.3 ± 1.5	Cycling	Pre-season	122	8.5 ± 0.4	18.3 ± 12.6	–	89.6 ± 4.7	41.0 ± 22.2	–	–	–	–
Suppiah (2015)[90]	4	M	-	Bowling	NA	8	6.5 ± 0.3	–	–	–	4.4 ± 1.5	106.4 ± 16.5	191.0 ± 7.4[b]		93.1 ± 7

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	4			Badminton	Heavy training phase	8	6.7 ± 0.3	–	–	–	1.0 ± 0.4	105.5 ± 19.5	157.6 ± 25.4 [b]		139.3 ± 19.6
Suppiah (2016)[50]	6	M	14.7 ± 1.3	Shooting	NA	6	6.4 ± 0.1	–	–	–	2.2 ± 1.7	100.8 ± 38.1	43.7 ± 6.8 [b]		29.0 ± 11.5
	8	M		Track and Field	Heavy training phase	8	6.4 ± 0.1	–	–	–	3.0 ± 2.4	102.9 ± 34.6	48.1 ± 6.6[b]		24.1 ± 3.8
Fuller (2017)[91]	21	M	22.5 ± 2.7	ARF & Rugby	Pre-season	67	7.6 ± 0.8	16.0 ± 15.5	–	88.7 ± 4.9	41.0 ± 21.6	–	–	–	–
Dunican (2017)[6]	25	M	25.0 ± 4.0	Rugby	Heavy training phase	25	6.6 ± 0.6	13.0 ± 11.0	–	89.0 ± 7.0	37.0 ± 30.0	–	–	–	–
Estivill-Domenech (2018)[35]	9	F	16.0 ± 0.8	Basketball	In season	9	6.7 ± 2.0	19.0 ± 13.6	–	95.2 ± 25.7	–	–	–	–	–
Gerber (2018)[30]	25	M/F	17.2 ± 1.6	NR	NR	26	7.0 ± 0.7	17.0 ± 18.0	–	91.3 ± 7.2	–	94.0 ± 21.0	14.0± 8.0	214 ± 47	95.0 ± 26.0
Knufinke (2018)[49]	98	M/F	18.8 ± 3.0	Multi-sports	In season	206	–	–	–	–	–	119.0 ± 42.0	234 ± 50[b]		97.0 ± 31.0
						12	8.1 ± 0.7	–	–	89.7 ± 7.2	22.9 ± 11.5	110.2 ± 15.9	28.4 ± 12.8	198.2 ± 38.1	147.6 ± 38.9
Romyn (2018)[25]	12	M	18.3 ± 1.0	Soccer	NR	12	8.2 ± 0.5 [c]	–	–	90.7 ± 5.7	25.3 ± 19.1	93.3 ± 29.1	28.3 ± 16.5	209.2 ± 46.4	158.8 ± 39.5
						12	8.0 ± 0.9 [c]	–	–	89.4 ± 10.6	34.5 ± 37.1	90.3 ± 42.7	29.5 ± 12.4	204.5 ± 42.4	158.5 ± 36.5

Notes: [a] Combination of N3 and N4 sleep stages as “deep sleep”; [b] Combination of N1 and N2 sleep stages as “light sleep”; [c] Included daytime naps in the results.

Abbreviations: ARF = Australian rules football; F = females ; M = males ; M / F = males and females ;NR = not reported; REM = rapid eye movement ; ROL = REM onset latency; SE = sleep efficiency; SOL = sleep onset latency;TST = total sleep time; WASO = wake after sleep onset.

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Table 3

Sleep-related data measured with actigraphy (mean ± SD).

First author (publication year)	*n*	Sex	Age (years)	Type of Sport	Training season	Nights Recorded	TST (h)	SOL (min)	SE (%)	WASO (minn)
Wall (2003)[53]	3/1	M/F	-	Swimming	Heavy training phase	8	–	–	81.6 ± 4.4	–
	4	F			Heavy training phase	8	–	–	94.6 ± 4.8	–
Richmond (2004)[77]	10	M	23.0 ± 2.0	ARF	In season	50	8.4 ± 0.1	–	88.0 ± 3.8	57.0 ± 18.0
Richmond (2007)[92]	19	M	24.1 ± 3.3	ARF	In season	76	8.9 ± 0.1	–	93.1 ± 1.0	–
Fietze (2009)[93]	9/15	M/F	27 ± 5.0	Dance	NA	168	7.0 ± 0.1	18.1 ± 9.5	81.1 ± 4.4	72.1 ± 27.7
						168	6.5 ± 0.1	15.4 ± 7.0	78.6 ± 4.7	74.2 ± 27.0
Leeder (2012)[9]	11	M/F	-	Canoeing	Competition phase	44	8.5 ± 0.1	19.1 ± 20.2	81.8 ± 4.3	66.0 ± 17.0
	14			Diving		56	8.8 ± 0.1	21.0 ± 19.0	80.9 ± 5.3	77.0 ± 19.0
	10			Rowing		40	7.8 ± 0.1	10.2 ± 6.6	82.5 ± 8.3	68.0 ± 32.0
	11			Short-track speed skating		44	9.2 ± 0.2	21.1 ± 15.1	77.2 ± 7.1	98.0 ± 46.0
Sargent (2014)[40]	6/1	M/F	22.5 ± 1.7	Swimming	Heavy training phase	84	5.4 ± 1.3	40.8 ± 43.2	70.7 ± 15.1	17.6 ± 8.8
						14	7.1 ± 1.2	31.8 ± 21.6	77.2 ± 7.5	16.2 ± 7.7
Hoshikawa (2013)[80]	7	F	19.6 ± 0.8	Distance running	Heavy training phase	49	8.2 ± 1.1	6.1 ± 8.6	93.6 ± 5.6	24.9 ± 20.4
Lahart (2013)[27]	3	M	37.5 ± 4.1	Cycling	Competition phase	3	7.2 ± 0.3	19.0 ± 12.0	84.0 ± 2.0	–
					Ultra-endurance competition phase	16	2.2 ± 0.3	24.0 ± 4.0	83.0 ± 6.0	–
Roach (2013)[81]	14	M	15.6 ± 0.5	Soccer (U17)	Pre-season	84	7.0 ± 0.5	–	80.0 ± 3.0	–
	12		17.4 ± 1.2	Soccer (U20)		72	6.4 ± 1.0	–	80.0 ± 6.0	–
Robey (2013)[94]	12		18.5 ± 1.4	Soccer	In season	72	7.2 ± 0.1	20.6 ± 10.9	89.4 ± 5.8	15.2 ± 10.9
Fowler (2014)[95]	6	M	23.4 ± NR	ARF	In season	6	7.1 ± NR	10.8 ± NR	82.3 ± NR	–
						6	8.6 ± NR	25.3 ± NR	83.3 ± NR	–
						6	6.1 ± NR	17.0 ± NR	83.7 ± NR	–
						6	6.7 ± NR	11.5 ± NR	83.6 ± NR	–
						6	8.2 ± NR	23.2 ± NR	84.8 ± NR	–
Hausswirth (2014)[44]	9	M	37.0 ± 6.0	Triathlon	Tapering	63	6.5 ± 0.1	9.0 ± 9.0	88.3 ± 6.3	–
						63	6.7 ± 0.8	7.0 ± 7.0	89.0 ± 6.5	–
						63	6.8 ± 0.7	7.0 ± 7.0	88.4 ± 5.5	–

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					Tapering	63	7.4 ± 0.1	4.0 ± 4.0	91.0 ± 1.9	–
	9		35.0 ± 8.0		Heavy training phase	63	7.2 ± 0.7	4.0 ± 4.0	90.0 ± 1.9	–
					Tapering	63	7.1 ± 0.5	6.0 ± 5.0	89.3 ± 2.8	–
					Tapering	63	7.2 ± 0.1	4.0 ± 1.0	90.0 ± 1.3	–
	9		35.0 ± 5.0		Heavy training phase	63	6.6 ± 0.9	5.0 ± 2.0	88.4 ± 1.7	–
					Tapering	63	7.1 ± 0.8	4.0 ± 2.0	89.4 ± 3.2	–
Lastella (2015)[36]	57/9	M/F	21.8 ± 3.4	Multi-sports (Individual)	Competition phase	754	6.5 ± 1.1	22.0 ± 26.6	85.9 ± 6.1	–
	104/20		22.2 ± 3.0	Multi-sports (Team)		613	7.0 ± 1.2	16.3 ± 20.1	86.4 ± 4.8	–
						126	7.4 ± 0.6	28.0 ± 13.7	86.4 ± 0.4	–
						21	7.3 ± 0.8	17.7 ± 1.7	86.9 ± 5.6	–
Lastella (2015)[96]	21	M	19.9 ± 1.7	Cycling	Competition phase	21	6.9 ± 0.9	36.4 ± 2.7	85.3 ± 5.0	–
						21	6.5 ± 0.9	36.4 ± 2.9	84.9 ± 5.0	–
						21	6.7 ± 0.8	38.1 ± 4.6	85.8 ± 6.0	–
						21	6.9 ± 0.8	29.2 ± 2.3	81.7 ± 7.5	–
						16	6.3 ± 1.7	–	81.2 ± 6.7	–
Lastella (2014)[97]	16	M	18.8 ± 0.9	Football	Pre-season	16	7.0 ± 1.4	–	83.1 ± 7.0	–
						16	6.2 ± 1.4	–	83.3 ± 6.7	–
						16	7.1 ± 0.5	–	–	–
Fowler (2015)[98]	16	M	27.0 ± NR	ARF	Pre-season	16	4.9 ± 0.6	–	–	–
Killer (2017)[52]	10	M	25.0 ± 6.0	Cycling	NR	80	6.6 ± 0.2	–	80.9 ± NR	–
					Tapering	124	7.4 ± 0.8	–	86.6 ± 4.8	–
						142	7.2 ± 0.9	–	84.4 ± 5.3	–
Lastella (2015)[28]	21	M	22.2 ± 2.7	Cycling	Ultra-endurance competition phase	141	7.0 ± 0.9	–	84.9 ± 5.4	–
						144	7.1 ± 1	–	85.1 ± 5.9	–
					Tapering	209	7.1 ± 0.9	–	85.7 ± 5.4	–
Romyn (2015)[99]	8	F	19.6 ± 1.5	Netball	In season	56	8.2 ± 0.5	28.2 ± 26.2	85.2 ± 3.5	35.4 ± 7.6
Sargent (2014)[41]	46/24	M/F	20.3 ± 2.9	Multi-Sports	Heavy training phase	157	6.8 ± 0.1	–	85.3 ± 7.6	–
						769	6.5 ± 0.1	–	85.6 ± 7.1	–
Sargent (2016)[10]	16	M	19.3 ± 1.5	Cycling	Pre-season	128	7.6 ± 0.6	17.0 ± 14.0	85.0 ± 5.0	80.0 ± 28.8

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Schaal (2015)[100]	10	F	20.4 ± 0.4	Synchronized swimmers	Heavy training phase	70	7.2 ± 0.2	17.0 ± 2.0	84.7 ± 1.3	–
						70	6.9 ± 0.2	28.0 ± 6.0	82.7 ± 1.6	–
Shearer (2015)[101]	28	M	24.4 ± 2.9	Rugby	In season	112	7.1 ± 1.0	34.0 ± 40.0	79.0 ± 9.0	–
						112	7.6 ± 1.2	28.0 ± 25.0	78.3 ± 11.3	–
						112	6.0 ± 1.5	38.0 ± 34.0	74.7 ± 11.1	–
Suppiah (2015)[90]	6	M	14.8 ± 0.9	Bowling	NA	30	6.0 ± 0.1	6.7 ± 10.8	–	38.7 ± 18.1
	5			Badminton	Heavy training phase	25	6.2 ± 0.1	3.4 ± 1.2	–	36.0 ± 15.1
Van Maanen (2015)[29]	9/1	M/F	42.9 ± 8.0	Roparum (8 runners & 2 cyclists)	Ultra-endurance competition phase	20	7.7 ± 2.0	–	–	–
Eagles (2014)[102]	10	M	24.3 ± 2.6	Rugby	In season	120	7.7 ± 0.1	–	87.0 ± 2.0	–
Fullagar (2016)[20]	14	M	-	Soccer	Pre-season	14	6.9 ± 1.1	12.6 ± 11.6	89.2 ± 4.0	29.6 ± 17.0
						14	6.2 ± 0.7	21.1 ± 16.9	84.6 ± 9.0	38.9 ± 27.5
						14	7.0 ± 1.1	9.8 ± 15.3	91.2 ± 4.1	25.2 ± 14.7
Louis (2016)[103]	10	M	31.0 ± 4.7	Triathlon	Heavy training phase	210	6.5 ± 0.1	10.0 ± 7.0	82.5 ± 6.8	–
Pitchford (2017)[104]	19	M	22.1 ± 3.5	ARF	Pre-season	152	7.0 ± 0.1	–	84.7 ± 6.5	68.9 ± 29.9
						152	7.0 ± 0.1	–	78.7 ± 6.5	96.5 ± 33.7
Sargent (2016)[73]	22	M	22.1 ± 2.7	ARF	Pre-season	308	8.1 ± 0.8	17.9 ± 11.1	82.2 ± 3.8	–
						308	7.7 ± 0.8	10.9 ± 13.3	83.5 ± 6.1	–
Suppiah (2016)[50]	29	M	14.7 ± 1.3	15 Shooting / 14 Track & Field	NA	145	5.5 ± 0.1	–	80.9 ± 8.6	74.7 ± 22.6
						58	6.8 ± 0.1	–	80.9 ± 6.9	86.7 ± 40.8
Thornton (2017)[42]	31	M	24.5 ± 3.9	Rugby	Pre-season	217	7.3 ± 0.1	21.0 ± 19.0	88.1 ± 4.2	42.2 ± 17.5
						403	5.9 ± 0.1	9.0 ± 7.0	79.6 ± 8.3	81.6 ± 39.9
Van Ryswyk (2017)[105]	25	M	23.7 ± 2.0	ARF	Pre-season	100	7.1 ± 0.9	–	82.3 ± 6.5	–
Vitale (2017)[82]	12	M	21.1 ± 2.3	Soccer (M-TYPE)	NR	12	–	10.6 ± 11.1	88.1 ± 5.7	–
						12	–	19.0 ± 17.8	78.3 ± 7.4	–
						12	–	11.9 ± 13.3	84.1 ± 7.7	–
	11	M	22.9 ± 2.9	Soccer (E-TYPE)		11	–	5.6 ± 2.4	87.2 ± 4.9	–
						11	–	13.4 ± 15.9	85.8 ± 5.5	–
						11	–	16.3 ± 16.2	83.3 ± 7.8	–
Caia (2017)[43]	7	M	24.3 ± 2.1	Rugby	Pre-season	49	6.9 ± 0.4	20.1 ± 14	87.3 ± 3.0	–
					In season	49	6.9 ± 0.4	17.0 ± 7.5	85.9 ± 1.8	–

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	15		25.5 ± 3.7			49	7.5 ± 0.1	16.0 ± 18.0	87.8 ± 3.8	–
Caia (2017)[39]	15	M	22.4 ± 2.4	Rugby	In season	49	6.8 ± 0.1	20.0 ± 18.0	84.5 ± 5.8	–
	15		18.8 ± 0.9			49	7.3 ± 0.1	24.0 ± 27.0	85.7 ± 6.6	–
Caia (2018)[5]	63	M	-	Rugby	Pre-season	264	6.9 ± 0.1	–	–	–
Choi (2018)[33]	6	M/F	8.2 ± 0.5	Badminton	Heavy training phase	42	6.9 ± 0.2	13.0 ± 1.0	80.0 ± 2.9	77.0 ± 9.0
	7		7.9 ± 0.4			49	6.0 ± 0.1	18.0 ± 1.0	67.0 ± 2.0	142.0 ± 8.0
Driller (2017)[38]	8/11	M/F	20.0 ± 2.0	Multi-sports	Pre-season	133	7.6 ± 0.8	–	92.3 ± 1.4	42.0 ± 29.0
Dunican (2017)[6]	9	M	17.2 ± 5.1	Judo	Pre-season	18	6.7 ± 0.8	29.5 ± 27.2	84.0 ± 9.3	7.5 ± 5
	1/8	M/F	18.9 ± 2.9			18	7.3 ± 0.8	46.0 ± 44.6	85.5 ± 5.6	6.5 ± 5
Juliff (2018)[106]	12	F	19.2 ± 0.9	Netball	In season	12	–	14.5 ± 15.1	–	–
						12	–	24.4 ± 27.4	–	–
Juliff (2018)[48]	42	F	19.2 ± 1.0	Netball	In season	504	7.3 ± 1.0	7.6 ± 3.8	88.3 ± 3.5	–
						252	8.0 ± 0.6	12.1 ± 6.2	89.3 ± 2.9	–
						504	7.3 ± 0.9	5.9 ± 5.5	87.3 ± 4.3	–
						252	7.0 ± 0.5	9.3 ± 9.0	88.0 ± 5.7	–
	16			ARF		112	6.8 ± 1.2	18.0 ± 22.0	85.5 ± 5.7	70.0 ± NR
Miller (2017)[107]	7	M	27.3 ± 3.4	Soccer	NR	49	6.7 ± 2.0	10.0 ± 15.0	85.5 ± 5.7	56.0 ± NR
	28			Rugby union		196	7.2 ± 1.6	9.0 ± 11.0	88.5 ± 4.2	57.0 ± NR
O'Donnell (2017)[108]	26	F	23.0 ± 6.0	Netball	Pre-season	182	7.3 ± 0.1	28.6 ± 15.6	80.6 ± 6.5	–
Staunton (2017)[109]	17	F	-	Basketball	In season	323	7.5 ± 1.5	–	92.0 ± 3.0	–
						323	7.7 ± 1.7	–	91.0 ± 4.0	–
						323	7.4 ± 1.8	–	92.0 ± 4.0	–
						323	8.3 ± 1.6	–	93.0 ± 4.0	–
Whitworth-Turner (2017)[110]	11	M	18.0 ± 1.0	Soccer	In season	54	8.1 ± 1.1	24.0 ± 15.0	94.0 ± 3.0	6.0 ± 9.0
Blanchfield (2018)[65]	11	M	35.0 ± 12.0	Running	Heavy training phase	44	7.0 ± 0.7	–	–	–
						44	6.7 ± 1.0	–	–	–
						11	6.9 ± 0.6	18.0 ± 15.0	93.0 ± 4.0	–
						11	7.0 ± 1.1	22.0 ± 22.0	92.0 ± 6.0	–
Caia (2018)[62]	24	M	25.4 ± 3.3	Rugby	In season	288	7.3 ± 0.8	9.3 ± 9.0	89.6 ± 2.9	–
						288	7.8 ± 0.3	11.1 ± 11.3	87.7 ± 3.9	–

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Carrico (2018)[111]	25	M	26.3 ± 4.7	Soccer	In season	95	6.6 ± 0.1	24.0 ± 9.0	85.0 ± 5.0	30.0 ± 16.0
						75	5.6 ± 0.1	35.0 ± 28.0	78.0 ± 7.0	29.0 ± 13.0
Costa (2019)[112]	16	F	20.4 ± 2.1	Soccer	In-season	368	7.3 ± 0.8	6.0 ± 4.0	88.0 ± 5.0	56.0 ± 30.0
						368	7.9 ± 0.7	4.0 ± 4.0	89.0 ± 4.0	55.0 ± 24.0
Costa (2019)[113]	17	F	21.6 ± 2.3	Soccer	In-season	306	7. 2 ± 0.5	8.0 ± 2.0	90.0 ± 4.0	37.0 ± 12.0
						51	9.0 ± 0.9	9.0 ± 2.0	89.0 ± 3.0	40.0 ± 10.0
						306	8.6 ± 0.6	4.0 ± 2.0	91.0 ± 3.0	44.0 ± 14.0
Dunican (2019)[114]	20	M	26.0 ± 3.0	Rugby	In season	20	6.4 ± 0.9	55.4 ± 4.4	78.7 ± 9.8	10.1 ± 8.0
						20	7.3 ± 1.3	44.8 ± 4.6	78.2 ± 10.3	10.9 ± 5.1
						20	7.8 ± 1.6	27.1 ± 4.8	83.0 ± 9.8	10.3 ± 7.7
						20	4.4 ± 2.7	28.1 ± 4.6	62.3 ± 30	9.2 ± 7.9
						20	6.7 ± 1.3	45.1 ± 4.8	80.8 ± 9.8	19.2 ± 23.7
						20	6.5 ± 1.1	31.2 ± 4.8	80.3 ± 12.5	10.7 ± 11.6
						20	7.3 ± 1.2	–	83.3 ± 7.5	10.6 ± 8.7
Knufinke (2018)[49]	42/56	M/F	18.8 ± 3.0	Multi-sports	In-season	282	7.5 ± 1.1	14.0 ± 17.0	88.7 ± 5.6	31.0 ± 16.0
Lalor (2018)[115]	45	M	22.1 ± 3.3	ARF	Pre-season	450	9.2 ± 0.1	12.0 ± 0.9	81.4 ± 0.4	79.0 ± 1.7
					In season	450	8.1 ± 0.1	11.0 ± 1.3	85.6 ± 0.5	57.0 ± 2.3
Nedelec (2019)[116]	20	M	26.0 ± 4.6	Soccer	In-season	73	6.4 ± 0.9	38.0 ± 35.0	79.1 ± 10.5	–
O'Donnell (2018)[117]	10	F	23.0 ± 6.0	Netball	In season	10	8.8 ± 0.1	27.5 ± 34.7	85.3 ± 7.2	–
						10	8.0 ± 0.1	38.5 ± 29.3	82.1 ± 8.9	–
						10	6.1 ± 0.1	67.0 ± 51.9	74.4 ± 10.1	–
O'Donnell (2018)[118]	11	F	23.0 ± 4.0	Netball	In season	11	8.5 ± 1.0	23.4 ± 14.7	82.4 ± 6.4	–
						11	6.8 ± 0.8	21.7 ± 25.6	79.4 ± 6.1	–
						11	7.4 ± 0.3	33.1 ± 16.3	84.1 ± 6.7	–
Rosa (2018)[59]	11/11	M/F	24.8 ± 3.4	Swimming	Pre-season	44	7.6 ± 0.6	21.0 ± 6.0	84.7 ± 0.7	49.0 ± 5.0
Thornton (2018)[119]	14	M	26.1 ± 2.9	Rugby	Pre-season	585	6.9 ± 1.2	–	92.3 ± 3.0	35.0 ± 15.6
Vitale (2019)[17]	12/12	M/F	26.0 ± 3.4	Volleyball	In season	24	–	9.0 ± 5.1	81.1 ± 4.3	12.2 ± 2.7
						24	–	12.3 ± 4.4	80.7 ± 7.9	16.4 ± 3.6
						24	–	6.8 ± 3.1	86.7 ± 4.1	11.1 ± 2.9
Lastella (2019)[120]	7	M	25.2 ± 3.2	Soccer	In season	63	7.0 ± 1.6	8.5 ± 11.1	87.5 ± 4.0	–
Ramos-Campo (2019)[121]	14	M	27.6 ± 7.4	Distance running	Heavy training phase	14	6.8 ± 1.9	–	93.5 ± 4.0	–

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						14	6.8 ± 1.1	–	88.2 ± 3.5	–
						14	6.2 ± 1.5	–	91.2 ± 5.2	–
						14	6.1 ± 1.2	–	84.3 ± 5.6	–
						70	6.7 ± 0.2	–	88.0 ± 0.0	–
						70	6.6 ± 0.2	–	89.0 ± 0.0	–
						70	7.0 ± 0.1	–	89.0 ± 0.0	–
						70	6.8 ± 0.3	–	88.0 ± 0.0	–
						70	7.2 ± 0.4	–	90.0 ± 0.0	–
Sekiguchi (2019)[58]	10	F	19.0 ± 1.0	Running	Competition phase	70	6.9 ± 0.1	–	90.0 ± 0.0	–
						70	6.9 ± 0.2	–	89.0 ± 0.0	–
						70	6.8 ± 0.4	–	88.0 ± 0.0	–
						70	6.7 ± 0.4	–	89.0 ± 0.0	–
						70	6.9 ± 3.0	–	89.0 ± 0.0	–
						70	7.1 ± 0.2	–	92.0 ± 0.0	–
						70	7.0 ± 0.3	–	87.0 ± 0.0	–
Suppiah (2018)[64]	12	M	13.8 ± 1.0	Shooting	NR	12	5.4 ± 0.6	–	82.0 ± 9.2	70.6 ± 41.6
						12	5.6 ± 0.2	–	84.4 ± 7.6	60.5 ± 34.0
	19		14.8 ± 1.0	Track & field	Heavy training phase	19	5.4 ± 1.0	–	79.0 ± 8.0	80.3 ± 34.3
						19	5.5 ± 1.0	–	80.6 ± 9.3	78.4 ± 41.4
Walsh (2019)[122]	9/3	M/F	21.0 ± 2.0	Swimming	Heavy training phase	168	7.7 ± 0.7	19.0 ± 12.0	85.0 ± 5.0	–
					Tapering	168	7.9 ± 0.6	19.0 ± 6.0	85.0 ± 5.0	–
						46	8.9 ± 0.8	17.0 ± 9.0	82.0 ± 5.0	–
					Competition	67	7.9 ± 0.7	27.0 ± 11.0	82.0 ± 3.0	–

Abbreviations: ARF = Australian Rules Football; F = females; M = males; M / F = males and females ;NR = not reported; SE = sleep efficiecy;SOL = sleep onset latency ;TST = total sleep time ; WASO = wake after sleep onset .

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Table 4

Overall sleep data calculations (mean ± SD).

Categorization		Group/ Phase/ Sports	TST (h)	SOL (min)	SE (%)	WASO (min)	ROL (min)	REM (min)	N1 (min)	N2 (min)	N3 (min)
According to monitoring tool		Overall	7.2 ± 1.1	14.8 ± 17.0	86.3 ± 6.8	52.7 ± 32.0	81.1 ± 31.2	114.1 ± 35.4	47.9 ± 31.0	240.5 ± 53.6	91.5 ± 35.8
		Polysomnography/Electroencephalography	7.6 ± 1.1	17.5 ± 17.6	90.2 ± 6.8	30.3 ± 25.3	–	–	–	–	–
		Actigraphy	7.2 ± 1.1	14.7 ± 16.9	86.3 ± 6.8	54.0 ± 31.9	–	NA	NA	NA	NA
According to sex		Males	7.2 ± 1.1	15.7 ± 17.0	85.2 ± 7.2	54.8 ± 32.9	84.9 ± 32.3	111.3 ± 26.0	55.8 ± 29.0	246.7 ± 53.3	82.9 ± 42.0
		Females	7.5 ± 1.2	9.7 ± 11.2	89.1 ± 4.8	46.6 ± 23.3	66.1 ± 23.5[a]	–	–	NA	NA
According to sex and monitoring tool	Polysomnography/Electroencephalography	Males	7.7 ± 1.1	17.7 ± 18.2	89.8 ± 5.5	32.3 ± 25.5	84.9 ± 32.3	111.3 ± 26.0	55.8 ± 29.0	246.7 ± 53.3	82.9 ± 42.0
		Females	7.4 ± 1.3	18.0 ± 8.8	95.2 ± 25.7[a]	7.5 a ± 9.1	66.1 ± 23.5[a]	–	–	–	–
	Actigraphy	Males	7.2 ± 1.1	15.5 ± 16.9	85.1 ± 7.2	57.0 ± 32.7	–	NA	NA	NA	NA
		Females	7.5 ± 1.2	9.6 ± 11.2	89.1 ± 4.7	47.1 ± 23.0	–	NA	NA	NA	NA
According to athletic expertise		Elite	7.2 ± 1.2	18.4 ± 19.4	85.9 ± 7.0	53.2 ± 32.2	–	116.2 ± 40.9	14.0 ± 8.0[a]	214.0 ± 47.0[a]	96.8 ± 30.4
		Semi-Elite	7.3 ± 1.0	9.5 ± 10.5	87.4 ± 6.1	52.1 ± 31.8	81.1 ± 31.2	111.3 ± 26.0	55.8 ± 29.0	246.7 ± 53.3	82.9 ± 42.0
According to sex and athletic expertise	Elite	Males	7.3 ± 1.2	17.1 ± 17.7	84.9 ± 7.2	56.0 ± 31.7	–	–	–	–	–
		Females	7.4 ± 1.4	27.2 ± 18.1	89.5 ± 5.3	–	–	–	–	–	–
	Semi-elite	Males	7.0 ± 1.0	11.4 ± 13.5	86.8 ± 6.8	50.1 ± 37.0	84.9 ± 32.3	111.3 ± 26.0	55.8 ± 29.0	246.7 ± 53.3	82.9 ± 42.0
		Females	7.6 ± 0.9	7.4 ± 7.4	88.7 ± 4.3	46.6 ± 23.3	66.1 ± 23.5[a]	–	–	–	–
According to age group		Childhood and	6.3 ± 0.8	14.5 ± 13.2	80.3 ± 8.8	74.0 ± 40.8	–	98.8 ± 26.2	17.3 ± 10.9	214.6 ± 42.2	82.7 ± 43.3

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		adolescence (6–17)									
		Early adulthood (18–24)	7.3 ± 1.1	15.1 ± 17.5	85.9 ± 6.4	53.7 ± 30.2	80.9 ± 32.0	116.8 ± 37.1	59.5 ± 29.1	254.0 ± 56.2	93.5 ± 33.9
		Middle adulthood (25–40)	6.9 ± 1.0	13.8 ± 15.5	86.9 ± 7.2	40.4 ± 28.0	82.7 ± 26.5[a]	–	–	–	–
According to age group and athletic expertise	Childhood and Adolescence (6–17)	Elite	6.4 ± 1.0	17.5 ± 16.8	82.0 ± 8.7	74.0 ± 37.8[a]	–	94.0 ± 21.0[a]	14.0 ± 8.0[a]	214.0 ± 47.0[a]	95.0 ± 26.0[a]
		Semi-elite	6.1 ± 0.7	13.9 ± 12.3	79.0 ± 8.7	74.0 ± 41.3	–	104.1 ± 30.5	26.0 ± 13.0[a]	216.0 ± 28.0[a]	69.5 ± 54.0
	Early adulthood (18–24)	Elite	7.2 ± 1.1	18.9 ± 19.6	84.7 ± 6.6	60.6 ± 31.2	–	119.0 ± 42.0[a]	–	–	97.0 ± 31.0[a]
		Semi-elite	7.6 ± 0.9	9.1 ± 10.9	88.7 ± 4.7	45.5 ± 26.7	80.9 ± 32.0	112.9 ± 26.1	59.5 ± 29.1	254.0 ± 56.2	85.1 ± 38.9
	Middle adulthood (25–40)	Elite	7.0 ± 1.2	16.6 ± 18.6	87.6 ± 7.4	29.2 ± 17.5	–	–	–	–	–
		Semi-elite	6.8 ± 0.6	10.0 ± 8.7	85.5 ± 6.6	73.2 ± 27.3[a]	82.7 ± 26.4[a]	–	–	–	–
According to sports season		Heavy training phase	6.7 ± 0.8	16.0 ± 17.9	84.5 ± 8.4	50.3 ± 46.4	88.7 ± 24.7	105.6 ± 25.4	57.7 ± 20.2	216.2 ± 32.9	74.6 ± 41.5
		In-season / Competition phase	7.4 ± 1.2	14.8 ± 18.3	87.5 ± 6.2	43.9 ± 25.4	81.6 ± 26.8	118.5 ± 39.7	–	–	97.0 ± 31.0
		Pre-season / Tapering	7.3 ± 1.1	14.6 ± 13.3	85.3 ± 6.7	59.4 ± 33.2	60.7 ± 8.9[a]	107.0 ± 25.0[a]	26.0 ± 13.0[a]	216.0 ± 28.0[a]	130.0 ± 19.0[a]
According to sport type		Purely aerobic	7.2 ± 1.0	16.5 ± 15.6	86.4 ± 6.2	48.9 ± 36.6	82.3 ± 31.8	111.8 ± 24.5	62.8 ± 24.5	248.5 ± 55.3	78.2 ± 30.6
		Power / Explosiveness	8.7 ± 0.9	24.2 ± 31.3	80.9 ± 5.3[a]	77.0 ± 19.0[a]	91.0 ± 44.7[a]	103.0 ± 34.1[a]	17.0 ± 40.5[a]	271.0 ± 55.1[a]	61.0 ± 33.0
		Mixed sport	7.4 ± 1.2	13.3 ± 15.5	86.6 ± 7.0	52.6 ± 30.8	–	107.9 ± 19.6	27.3 ± 12.6	206.3 ± 34.3	139.5 ± 28.9
		Individual sports	7.0 ± 1.1	18.1 ± 20.4	86.1 ± 6.5	53.8 ± 38.2	84.0 ± 34.5	109.3 ± 27.5	65.8 ± 31.2	256.7 ± 55.8	66.9 ± 34.0
		Team sports	7.4 ± 1.2	13.5 ± 16.0	86.7 ± 6.8	51.6 ± 29.6	–	108.7 ± 20.1	27.3 ± 12.6	206.3 ± 34.3	139.6 ± 32.1

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Notes: [a] Data were available only from 1 study.

Abbreviations: NA = not applicable; NR = not reported; REM = rapid eye movement ; ROL = REM onset latency ; SE = sleep efficiency ;SOL = sleep onset latency;TST = total sleep time; WASO = wake after sleep onset.

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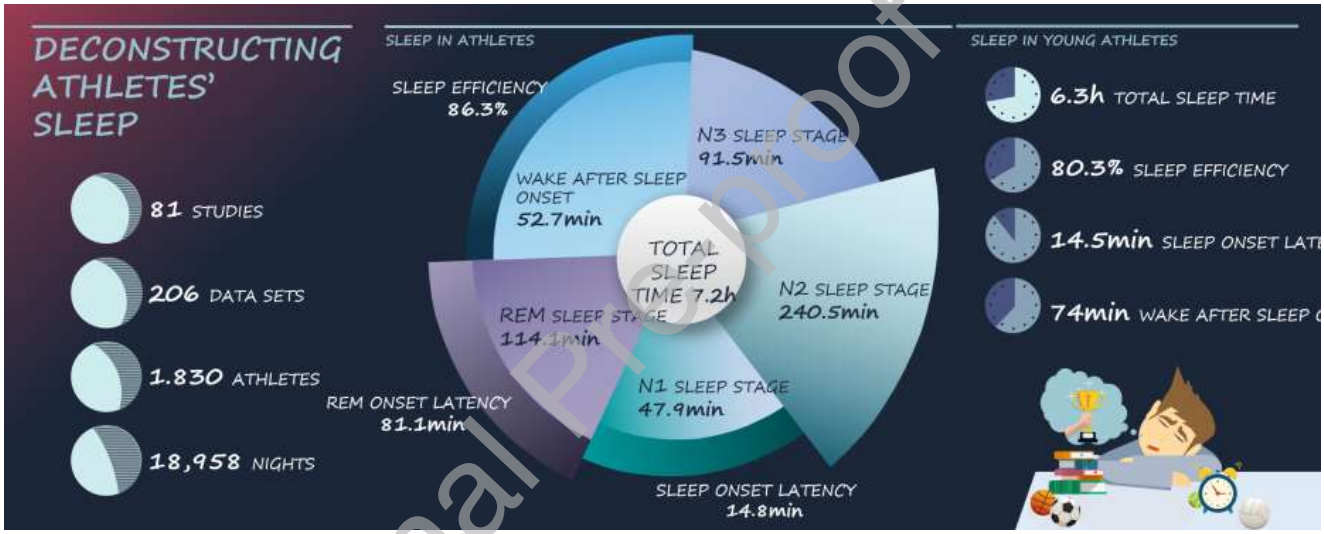
Graphical Abstract



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